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[4757]-1077

S.E. (Computer Engineering) (Second Semester) EXAMINATION, 2015

MICROPROCESSOR AND INTERFACING TECHNIQUES

(2012 Pattern)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Answer total *four* questions. Q. No. 1 *or* Q. No. 2, Q. No. 3 *or* Q. No. 4, Q. No. 5 *or* Q. No. 6, Q. No. 7 *or* Q. No. 8.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

1. (a) What are different components of MS-DOS ? Explain DOS loading with the help of neat diagram. [8]

(b) Compare 8086, 80386 and i7 processor on the basis of architectural features. [4]

Or

2. (a) Draw and explain block diagram of 8259APIC. [8]

(b) Write the initialization instructions of 8259A PIC, to meet the following specifications : [4]

(i) Interrupt type 32

P.T.O.

- (ii) Edge Triggered, single and ICW4 needed, interval of 8
- (iii) Mask IR1 & IR3 interrupts.

3. (a) Draw and explain I/O of BSR mode of 8255 with appropriate control word formats. [4]
- (b) Draw and discuss internal block diagram of 8251 USART. [6]
- (c) Define Resolution and Offset error terms of ADC. [2]

Or

4. (a) Design a control word format for square wave generator with 1ms period, the input frequency for 8253 is 1 MHz. [4]
- (b) Draw and explain the following 8279 commands : [4]
- (i) Keyboard/Display mode set command
 - (ii) Read FIFO/Sensor RAM command.
- (c) Explain with neat diagram sequence of DMA operation. [4]
5. (a) Draw and discuss the interface between 8086 and 8087. [7]
- (b) With proper timing diagram explain Read cycle in minimum mode of 8086 microprocessor. [6]

Or

6. (a) With the help of neat diagram explain minimum mode configuration of 8086. [6]

- (b) Draw and explain format of control and status word of 8087 NDP. [4]
- (c) Explain the following 8087 instructions with *one* example each : [3]
- (i) FSQRT
- (ii) FLDZ
- (iii) FADD
7. (a) Explain the features of 82801 IJR I/O Controller Hub. [5]
- (b) Draw and explain block diagram of X58 Chipset. [8]
- Or*
8. (a) Draw and explain block diagram of i5 motherboard. [8]
- (b) Write a short note on Intel's QPI Technology. [3]
- (c) Explain ICH10 PCI Interface. [2]